

STATES AND PHASES OF MATTER

Physical Properties of Liquid

1. Vapor pressure and Boiling point:

“The pressure exerted by vapors of liquid at its boiling point is called vapor pressure of liquid.”

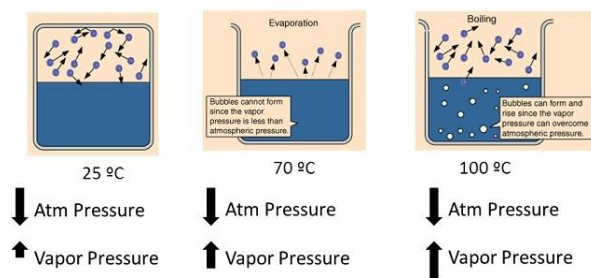
When a liquid is heated, its vapour pressure increases with increasing temperature due to the decrease of intermolecular forces. As a result, more and more vapor are released into the air. A stage is reached where the liquid begins to boil.

“The temperature at which the vapour pressure of a liquid is equal to the atmospheric pressure or some external pressure is called the boiling point of the liquid.”

e.g. Boiling Point of water at 760mmHg = 100 °C

Boiling Point of water at 23.7mmHg = 25 °C

Boiling Point



2. Evaporation:

“The spontaneous change of a liquid into its vapour at the surface of liquid at given temperature is called evaporation.”

The molecules in liquids are in motion and move within the liquid volume. The energy of molecules is not equally distributed.

The molecules having high kinetic energy move faster while other molecules having low kinetic energy move slowly.

When the high speed molecules reach the surface of the liquid it may escape the attraction of its side or neighboring molecules and leave the surface of the liquid.

Evaporation takes place at all temperatures. Evaporation causes cooling. When the high energy molecules leave the liquid during evaporation, low energy molecules are left behind, the average kinetic energy of

molecules of liquid decreases and temperature of the liquid falls so heat moves from surrounding to the liquid and temperature of surrounding falls.

Factor effecting the rate of evaporation:

Evaporation is controlled by different factors.

Surface area of a liquid:

When surface area of a liquid increases, evaporation increases because evaporation takes place from liquid surface. When liquid surface increases then more molecules can escape and rate of evaporation increases.

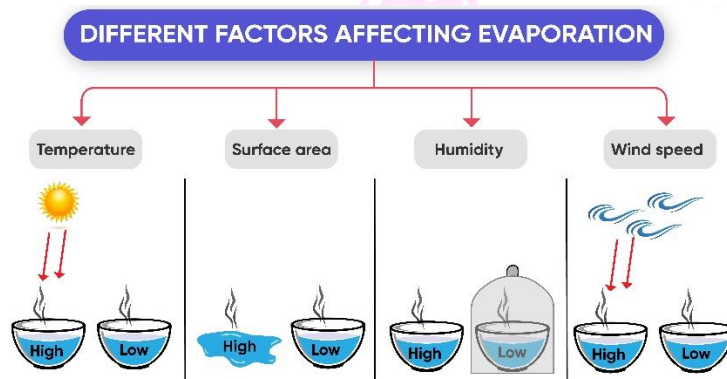
Intermolecular forces:

Intermolecular forces is another factor which affect evaporation. When intermolecular forces are strong molecules of liquid cannot leave the liquid surface easily and evaporation decreases. Rate of evaporation is slow when intermolecular forces are strong and when intermolecular forces are weak, the rate of evaporation is faster.

For example, Gasoline and ethyl alcohol whose molecules have weaker intermolecular forces (London forces of attraction) evaporation is much faster than water. Where molecules of water have strong intermolecular forces hydrogen bonding.

Temperature:

At high temperature molecules have greater energy and rate of evaporation increases.



3. Vapour Pressure:

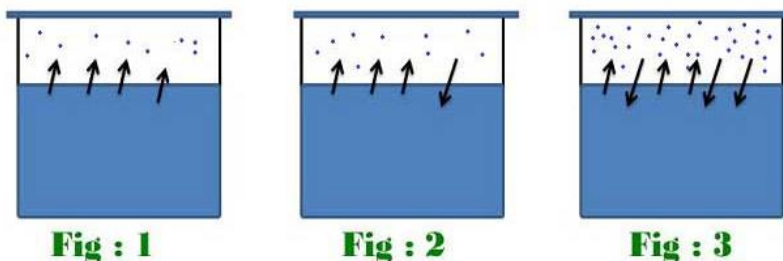
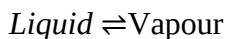
"The liquid molecules having high kinetic energy leave the liquid surface and escape out of liquid and this process is called evaporation."

When the surface of liquid is open the liquid molecules mixed up with air above the liquid and the molecules leave the surface and go away.

If the surface of liquid is covered and the system is closed the molecules of liquid evaporate and start gathering above the surface of liquid. These evaporated molecules collide with surface of liquid and also with walls of the container. These molecules are captured by surface of the liquid.

*“This coming back of vapours to liquid and captured by liquid surface is called **condensation**. ”*

*“Both the processes evaporation and condensation continue till a time reaches when the rate of evaporation is equal to the rate of condensation. This state is called the state of **dynamic equilibrium**. ”*



Vapour pressure of a liquid is the pressure exerted by vapour of the liquid in equilibrium with the liquid at a given temperature. At a constant temperature the number of molecules leaving the surface of liquid is equal to the number of molecules coming back into liquid.

At one moment the molecules of liquid are in liquid state may be in vapour state in next moment during evaporation.

Factor affecting the Vapour pressure:

The vapour pressure is independent of the amount of liquid or volume of container.

1. Surface area:

The surface area has no effect on vapour pressure. Larger surface area has larger target for evaporation and for returning the molecules.

2. Nature of liquid:

The vapour pressure of liquid depends on nature of the given liquid (intermolecular forces between its molecules, size of molecules whether small or large molecules)

3. Temperature:

Increasing temperature increases the kinetic energy of molecules. At high temperature molecules of liquid have high kinetic energy and capacity of molecules to break the intermolecular forces and escape to leave the liquid surface also increases and as a result the rate of evaporation increases this causes the increase of vapour pressure.

Water shows increase of vapour pressure from 4.579 torr at 0°C to 9.209 torr at 10°C, 527.8 torr at 90°C and 760 torr at 100°C

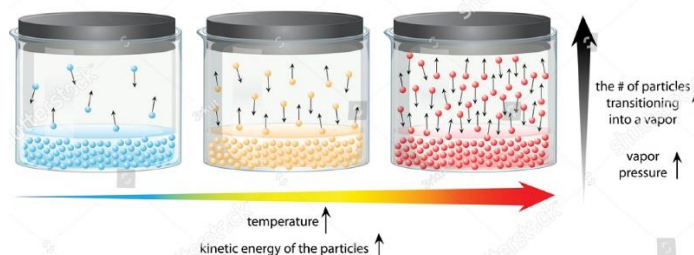
4. Intermolecular forces:

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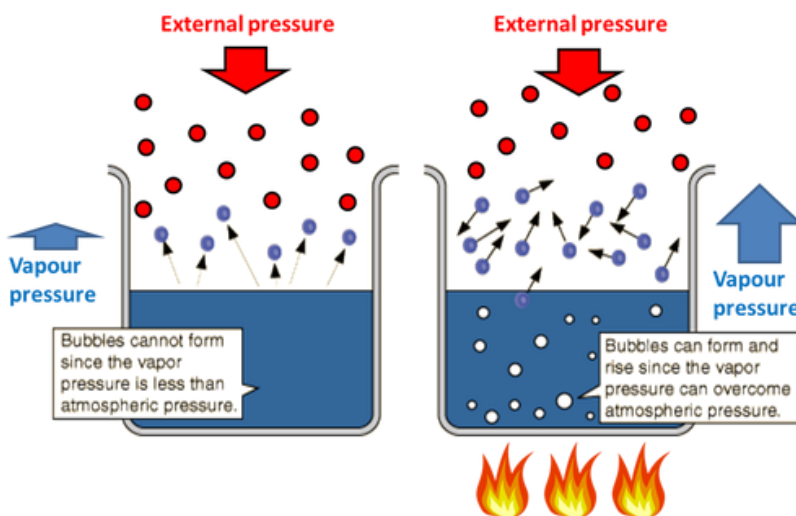
Intermolecular forces are also a factor which decides increase or decrease in vapour pressure of a liquid. At a particular temperature stronger the intermolecular forces lower the vapour pressure and weaker the intermolecular forces higher the vapour pressure.

At the same temperature different liquids have different vapour pressure.

At 20°C Glycerol has 0.00016 torr and Isopentane 580 torr vapour pressure.

**5. Effect of external atmospheric pressure on B.P:**

- ✓ Water boils at 100°C at 760 torr pressure at sea level and boils at 120°C at 1489 torr pressure. While boils at 25°C at 23.7 torr pressure. At Murree hills where external pressure is 700 torr water boils at 98°C while water boils at 69°C at the top of Mount Everest where external pressure at 323 torr.
- ✓ When the external pressure increases artificially the boiling temperature increases as in pressure cooker. The external pressure in pressure cooker is 2 atm so water boils at 120°C. A pressure cooker is a closed container from which water vapor cannot escape and pressure is created and the boiling temperature rises, therefore food is cooked quickly.
- ✓ In vacuum distillation liquid boils at low temperature which requires less amount of heat. Under this process boiling point of liquid is decreased and liquids like glycerine having boiling point 290°C at 760 torr pressure can decompose at this temperature and cannot be distilled at this temperature. By vacuum the boiling point of glycerine decreases to 210°C at 50 torr and is distilled without decomposition.



4. Boiling Point:

“The temperature at which vapour pressure of liquid become equal to the external atmospheric pressure is called boiling point.”

OR

“A temperature at which liquid converts into vapour is called boiling point.”

Explanation:

By heating the liquid, its vapour pressure increases. When heating is continued a stage will reach when the vapour pressure of the liquid becomes equal to external pressure or atmosphere pressure. *The temperature at which the vapour pressure of a liquid becomes equal to external pressure or atmospheric pressure is called boiling point or boiling temperature.*

At this temperature boiling starts. The kinetic energy of molecules increases by heating and temperature of the liquid also increases. When the kinetic energy of the liquid molecules become maximum then further heat supplied will not increase the temperature (temperature remains constant at boiling point) because the further heating will break the intermolecular forces and the liquid is converted into vapours.

5. Viscosity:

“The resistance to flow of a liquid is called viscosity.”

The higher the viscosity, the slower the liquid flows. Viscosity measures how easily molecules slide past each other.

- ✓ A common observation is that water can be poured from one container to another very quickly compared to honey and glycerin.
- ✓ To understand viscosity, think of a fluid flowing in a pipe as consisting of a series of concentric circular layers.
- ✓ Resistance is caused by internal friction between layers of molecules. The layer next to the walls have the lowest speed. Each layer attracts the other and thus causes resistance to flow.

Units:

SI units of viscosity is Pascal Second (Pa.s). Non-SI unit of viscosity is poise.

1 poise = 0.1 kg m's⁻¹ or gcm's⁻¹

1Pa.s 1 kgm's⁻¹ = 10 poise

Factor Affecting the viscosity of liquid:**1. Size and shape:**

The size and shape of the molecule strongly influences viscosity. Liquids such as water, acetone, benzene and methanol, whose molecules are small and compact, have a low viscosity. In liquids with large and

irregular molecules, such as honey, glycerin tends to tangle with each other. This hinders the flow of molecules and leads to high viscosity.

2. Intermolecular force

The stronger the intermolecular force, the higher the viscosity. Liquids whose molecules form hydrogen bonds are more viscous than others without hydrogen bonds. For example, water is more viscous than methanol mainly due to extensive hydrogen bonding.

3. Temperature:

Molecules move faster when the temperature rises. This is because; an increase in temperature decreases intermolecular forces. This dependence is quite visible for very viscous liquids such as honey and syrup. These liquids are easier to pour when hot than cold.

6. Surface tension:

"A force in dynes acting at right angles to a unit length of liquid surface is called surface tension."

Surface tension can also be defined as

"The amount of energy required to expand the surface of a liquid per unit area."

Units

SI unit of surface tension is joule per per square meter, J/m^2 or Newton per meter, N/m .

Explanation:

Surface tension is the property of the surface of liquids to behave as if a membrane were stretched over it. All molecules below the surface of a liquid are surrounded by other molecules in all directions. Thus, the force exerted by such molecules is balanced in all directions, while at the surface of a liquid there are molecules beside and below but none above it. This results in an unbalanced force that pulls the surface molecules inward. Molecules on the surface therefore feel an inward pull, creating surface tension.

Factor Affecting the Surface tension of Liquid:

Surface area:

For a molecule to reach the surface, it must overcome the downward force of attraction. This means that work must be done to bring it to the surface. Therefore, increasing the surface area of the liquid requires input energy.

Molecules on the surface of a liquid are less stable than inside it, so a liquid is stable if it has the fewest molecules on its surface. This happens when the surface of the liquid is the smallest.

Spheres have less surface area per unit volume than anything else. Therefore, small liquid droplets are usually spherical.

Effect of intermolecular forces:

The surface tension of a liquid directly depends on the strength of the intermolecular forces. The stronger the intermolecular forces between the liquid molecules, the higher the surface tension and vice versa.

For example, water has a higher surface tension than many other liquids, such as alcohols, ethers, benzene, etc. This is due to the strong hydrogen bonding between water molecules.

Effect of temperature:

The surface tension of a liquid decreases with increasing temperature. This is because the increased kinetic energy of the molecules reduces the strength of the intermolecular forces. It is different for different liquids because of different types of intermolecular forces

Concept of hydrogen bonding to explain the properties of water

Hydrogen bonding in water contributes to its unique properties, which are as follows.

1. Surface tension:

Surface tension is affected by strong intermolecular attractive forces. Water has hydrogen bonding which is strong intermolecular force. This creates a high surface tension.

2. Vapour pressure:

Vapour pressure of a liquid is affected by intermolecular attractive forces. Strong forces of attraction make the evaporation slow. Due to hydrogen bonding in water which is strong intermolecular attractive force decreases evaporation and so water has low vapour pressure.

3. High heat of vaporization of water:

Heat of vaporization of water compared to other liquids is high because of strong intermolecular forces of hydrogen bonding in water. More heat is required to overcome these attractive forces to vaporize water and as a result water show high heat of vaporization than other liquids.

4. High heat of vaporization of water results in high boiling point.

Due to strong hydrogen bonding water has low vapor pressure. So, more heat is required to enhance evaporation and produce vapour pressure equal to external pressure. As a result, water has high B.P.

Energetic of phase changes**Change in enthalpy (ΔH):**

The change in energy during chemical reaction at constant pressure is called change in enthalpy.

Unit: KJ/mol

Molar Heat of Fusion:

The amount of heat required to convert one mole of a solid into its liquid state at its melting point is called molar heat of fusion (ΔH_f).

Example

Molar heat of fusion for ice is $+6.02 \text{ kJ mole}^{-1}$

**Molar heat of vaporization (ΔH_v):**

The amount of heat required to convert one mole of a liquid into its vapours at its boiling point is called molar heat of vaporization.

Example**Energy changes and intermolecular forces:**

As a result of *melting of a solid*, a small change in intermolecular distance and potential energy takes place in atoms, molecules or ions.

On the other hand on evaporation of a liquid atoms, molecules or ions undergo large changes in their intermolecular distance and potential energy. Therefore, heat of vaporization is much greater than that heat of fusion.

Importance of heat of fusion in studying the Glaciers and Ice Caps

- ✓ Heat of fusion plays an important role to study behavior of glaciers, particularly polar ice sheets.
- ✓ The melting and freezing of ice are closely tied to the heat of fusion. Hence, it holds great significance to understand the response of glaciers and ice sheets to fluctuations in temperature.
- ✓ The Earth is getting hotter due to global warming. This means that the ice sheets at the North and South Poles are starting to melt. Scientists need to look at how much energy it takes to melt this ice, called the "heat of fusion". By understanding this, they can better predict how much the ice will melt and how that will affect sea levels and the environment.
- ✓ The quantity of water released during the melting of ice is directly proportional to the heat of fusion.
- ✓ Through the analysis of the heat absorbed during the melting process of ice cores, researchers are able to obtain valuable information about previous climatic conditions.

Different between Crystalline and Amorphous solid

Crystalline solid	Amorphous solid
They have definite shape and geometrical form	They don't have definite geometrical form
They have sharp melting points	They melt over a wide range of temperature.

They are rigid and incompressible.	They are also rigid and cannot be compressed, however graphite is soft because of its unusual structure.
They give a clean cleavage. i.e they break into pieces with the plane surfaces.	
They are anisotropic in nature	They are isotropic in nature.
They have definite heat of fusion.	They don't have definite heat of fusion.
Example: NaCl, diamond, Ice, glucose, ZnSO ₄	Example: Glass, plastic, Rubber

Properties of Crystalline solid

1. Geometrical shape:

Crystals have a geometric shape. The particles of solids (ions, atoms or molecules) are arranged in three-dimensional space.

The internal angles of a crystal (where surfaces intersect) are always the same, no matter in what form they are grown.

2. Melting point:

Crystalline solids have sharp melting points. When a solid is heated, its particles start vibrating at higher temperature and transfer their kinetic energy throughout the solid. At the melting point, their vibration energies become so high that they start leaving fixed positions simultaneously and the solid turn into a liquid.

3. Cleavage planes:

Crystalline solids break up along certain planes called cleavage planes. When pressure is applied to it along the cleavage plane, it changes into small crystals of the same size and shape as that of the original crystal. For a given crystalline solid, the cleavage planes are bent towards each other. The angles between them vary from one crystalline solid to another.

4. Habits of a crystal

"The shape of a crystal in which it usually grows is called habit of a crystal."

Crystals slowly form when a saturated solution is cooled. With slow cooling, liquid crystals are formed.

Crystals form by growing in different directions. If the growth conditions remain the same, crystals of the same shape will form.

For example, cubic crystals of sodium chloride are obtained from its aqueous solution. But if the conditions are changed, the shape of crystals also changes.

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For example, if 10% urea is present in aqueous solution of sodium chloride, octahedral crystals of sodium chloride are obtained.

For More Information:

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